

National Institute of Technology Kurukshetra
End Semester Examination, Even Semester 2025-26

Subject: Artificial Intelligence **Subject Code:** DSPC204 **Branch:** B.Tech. AI-DS
Semester: IV **Maximum Marks:** 50 **Date:** 05/08/2026 **Time:** 3 Hours
Roll No: _____

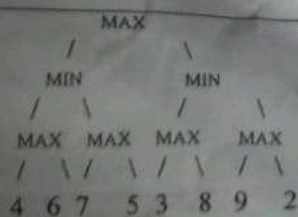
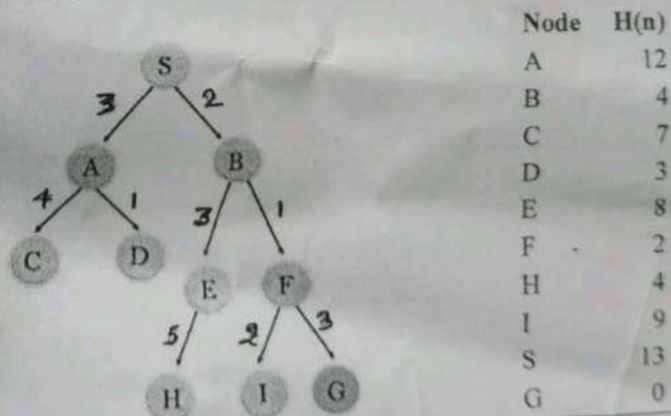
Instructions: There are 10 questions in total, and all are compulsory. Please read each question carefully before answering.

- 1 (a) Explain the following approaches to defining AI: (i) Thinking Humanly, (ii) Acting Humanly (iii), Thinking Rationally, and (iv) Acting Rationally
 (b) Compare these approaches in terms of objective, advantages, and limitations. And which approach is most suitable for designing modern AI systems?

- 2 A robot has to move from Room A to Room G inside a building. Define two formulations of the problem. Explain the four components of these two problem formulations. (i) Initial state, (ii) Goal state, (iii) Successor function and (iv) Path cost.
 Compare these two formulations. Which one is better, and why?

- 3 Compare BFS, DFS, uniform cost search, iterative deepening search, depth limited search based on following parameters in a table: (i) Completeness, (ii) Time complexity, (iii) Space complexity, and (iv) Optimality.
 Use b as the maximum branching factor, d as the depth of the shallowest solution and m as the maximum depth of the state space (may be infinite) to define time and space complexities.

- 4 Consider the following graph where S is the start node and G is the Goal node.
 (a) Apply the A* search algorithm to the given graph. Compute $f(n)=g(n)+h(n)$ for each expanded node. Determine the optimal path from the start node to the goal node and calculate its total path cost.
 (b) If the heuristic value of node F is changed from 2 to 8, would the optimal path found by A* necessarily change? Justify your answer.



- (a) Apply the Minimax algorithm.

(b) Calculate the utility value at every internal node.

(c) Identify the optimal move for MAX.

Consider the following 8-Queens configuration.

Column : 1 2 3 4 5 6 7 8

Row : 5 3 1 7 2 8 6 4

(a) Calculate the heuristic value (number of attacking queen pairs).

(b) Suggest one neighbouring state with a lower heuristic.

(c) Explain how Hill Climbing proceeds from this state.

Consider the Bayesian Network

(a) Explain the meaning of every dependency.

(b) Write the complete joint probability distribution.

(c) Explain conditional independence using d-separation.



For Bayesian Network in the above question, the following probabilities are given:

$P(B)=0.001$, $P(E)=0.002$, $P(A|B,E)=0.95$, $P(A|B,-E)=0.94$, $P(A|-B,E)=0.29$, $P(A|-B,-E)=0.001$, $P(J|A)=0.90$, and $P(J|-A)=0.05$ where B: Burglary, E: Earthquake, A: Alarm, J: JohnCalls.

Calculate $P(A)$ and $P(J)$, showing all intermediate calculations.

(a) Explain the Variable Elimination algorithm.

(b) Illustrate the procedure using a Bayesian Network of your choice.

(c) Why is Variable Elimination computationally more efficient than complete enumeration?

Why is an admissible heuristic important in A* search? Prove using one suitable example that an overestimating heuristic may fail to find the optimal solution.